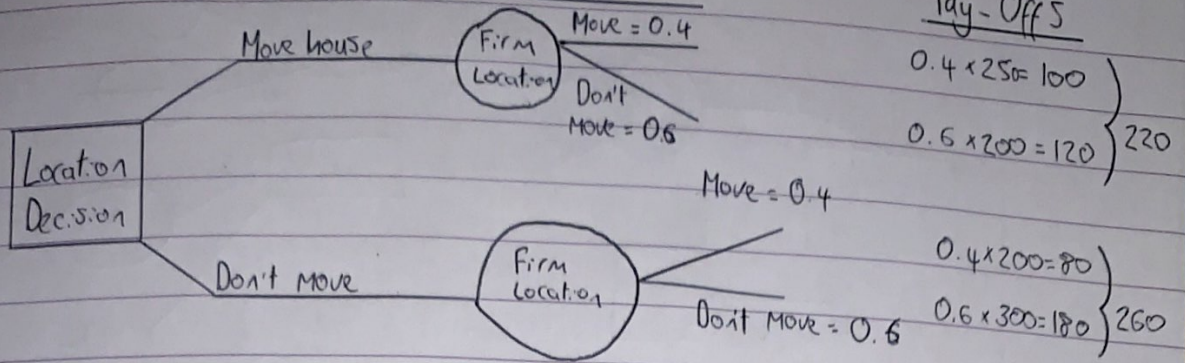


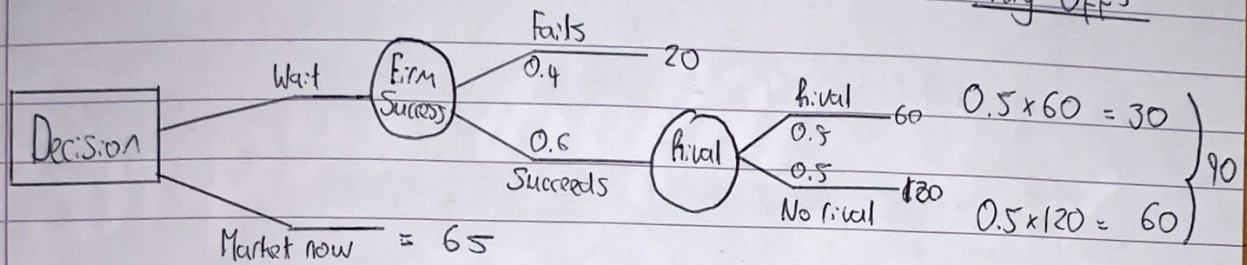
Supervision 2

1.



This is a cost minimisation decision, so they should move house as this has the lower EMV, £220 compared to £260

2.



Holling back:

$$90 \times 0.6 + 0.4 \times 20 = 62$$

$$\text{EMV Success} + \text{EMV failure} = 62$$

$62 < 65$ so better to market now

For wait to be more attractive, its $\text{EMV} > 62$

EMV of wait:

$$0.4(20) + 0.6(60p + 120(1-p))$$

→ This equation shows EMV of the "wait" half of the decision tree, allowing p to vary and holding pay-offs constant

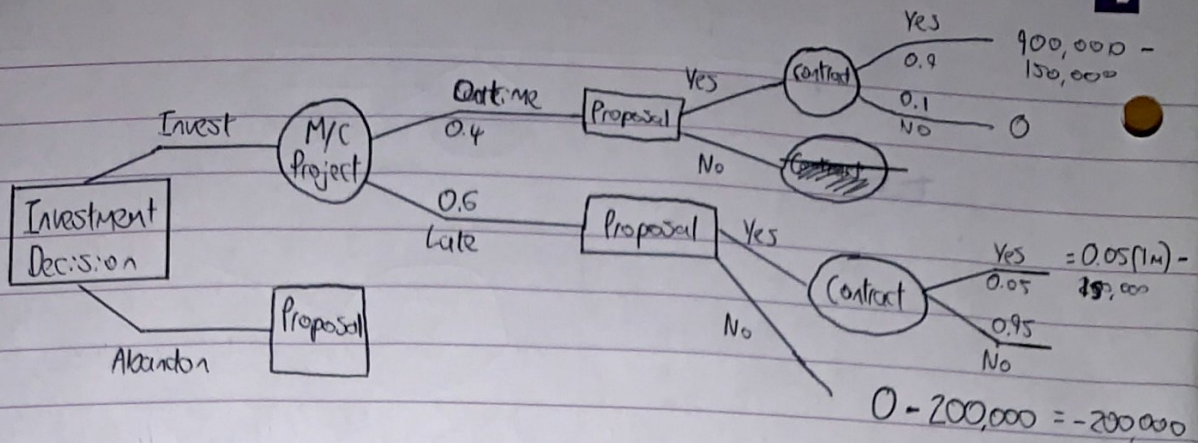
$$\text{EMV} = 8 + 0.6(120 - 60p)$$

$$= 8 + 72 - 36p$$

$$= 80 - 36p$$

To choose wait, $80 - 36p > 65$. Therefore when p , probability of a rival firm, < 0.41 , it is better to wait.

3.



EMVs

$$\text{Invest, late, Proposal} = -200,000 + 50,000 + (0.05(1,000,000) - 150,000) + 0$$

$$= -350,000$$

$$\text{Invest, late, } \overset{\text{No}}{\text{Proposal}} = -200,000$$

Therefore EMV for invest, late is -200,000

$$\text{Invest, on time, proposal, contract} = -200,000 - 50,000 + 900,000 - 150,000$$

$$\text{Invest, on time, proposal, No contract} = -250,000$$

Final answer was £86,000

Q4.

Q1. The flaw of averages is the idea that the output of a system, such as a decision tree, cannot be accurately predicted using averages of inputs which are not weighted by probabilities. For example, an explorer crossing a river needs to know the river's depth at each point, not merely the average. The average depth might be 3ft, but if a portion is 7ft he could be in danger.

Q2. Using decision trees, we calculate an expected market value for a decision or project. Monte Carlo simulation focuses not on the average NPV, but shows how the project's NPV varies with respect to the different possible input values, accounting for uncertainty.

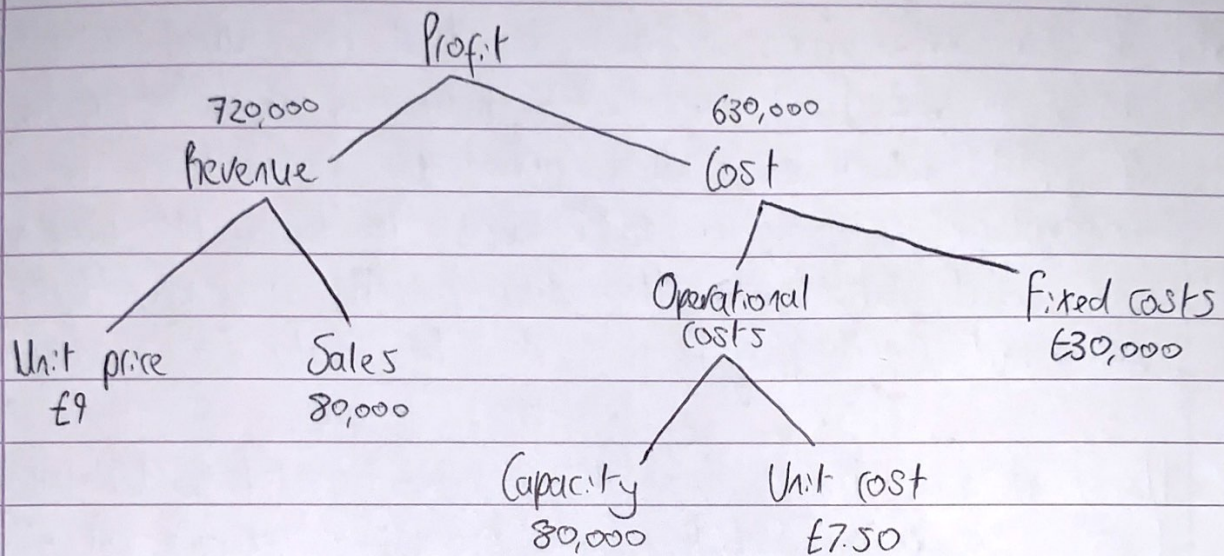
The method is a sampling technique which runs through our process (decision tree) and calculates how output varies with respect to a range of inputs.

Q3. The Static Model is a framework of analysis which relates a range of input data to an expected output. For example, it could relate the NPV of a project to the variables comprising the project's costs and revenues, through a decision tree.

The issue with the Static Model relates to potential lack of certainty with the input values, which could lead to incorrect NPV predictions. "What-if" analysis is sensitivity analysis which shows how changes to uncertain input values would affect output. For example, "what if" analysis could show us how NPV would change throughout a range of unit prices. Further, tornado diagrams conduct multiple sensitivity analyses and show which input's variability will have the greatest effect on NPV.

Q4.

(a)



Using this Static Model, which takes the average unit price, Sales and capacity, as well as expected unit cost, the expected profit is £90,000 ($£720,000 - £630,000$).